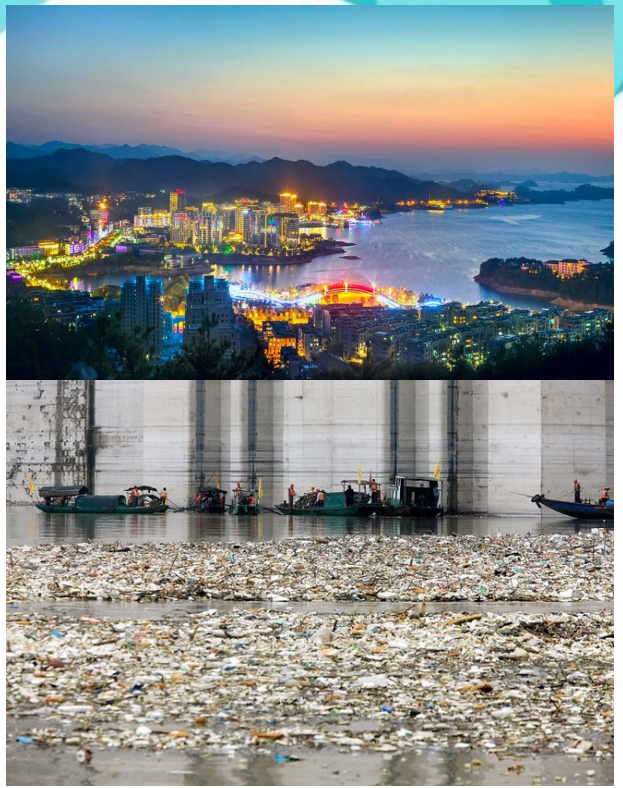
RIVER CLEANER



Senior Design Group 32

Instructor: **Dr. Jennifer Atchison**

Advisor: **Dr.B.C.Chang, Dr Najafi Ahmad**

Sponser: **HuaYuan Construction LLC**

Group members: ***Jie Xu,***
Tongming Huang, Zheng Yu, Xiao Tan, Chun Leung Devon Liu

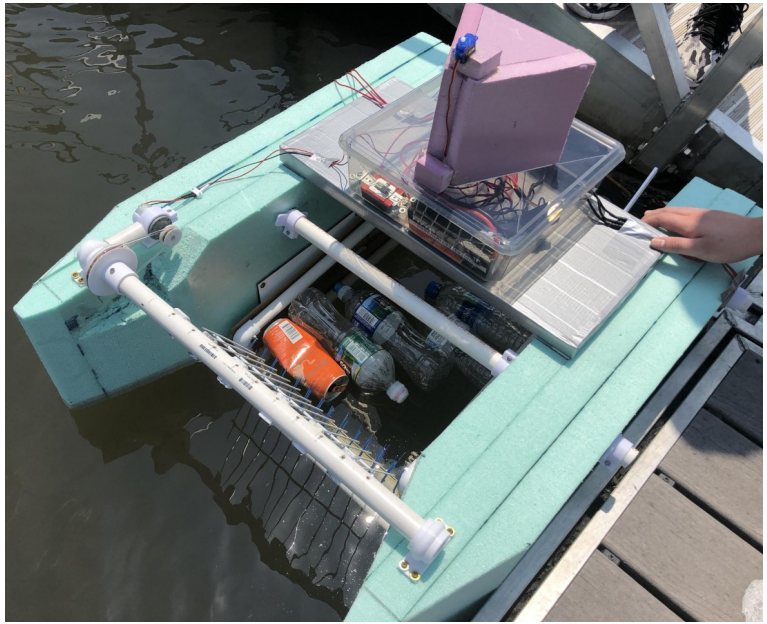
Presentation Date: 05/22/2019

Presentation Pathway

Intro	Prototype Manifestation	Results of This Term	The Future
● Motivation ● The Problem At Hand ● Our Solution	● The Process ● Improvements Made To The Original Idea ● Experiments Performed and Results	● Outcomes Found ● Cost Analysis ● Project Schedule ● Did this succeed (generally) ● Classes Utilized ● Research Resources	● Plan For Future ● What we Learned

PROBLEM SET

- *Task Assigned*:
  - Create a robot that can collect garbage in the local river.
- *River Info*:
  - Width of 4 meter and average water flow of 1.5 m/s
- *Garbage Info*:
  - Small bottles and plastic bags.
  - Will randomly flow down river.
- *Operator Info*:
  - Easy to operate.
  - Easy enough for workers to maintain.



STAKEHOLDER'S NEEDS

Needs	Numerical Value	Final Product Values
Efficiently collects trash	<30 seconds (10 meters distance between trash and robot)	about 25 seconds
Easy to maintain	< 1 hour of maintainance time per week	about 1 min. of maintenance
Within reasonable budget	$1000	$704.22
High garbage capacity on one charge	.05 m^3	.045 m^3
Can cruise at a stable velocity in the river	1-2 m/s	1-2 m/s
Working time for full charge	4 hours	about 3 hours
Easy movement on the river	near 0 turning radius	near 0 turning radius

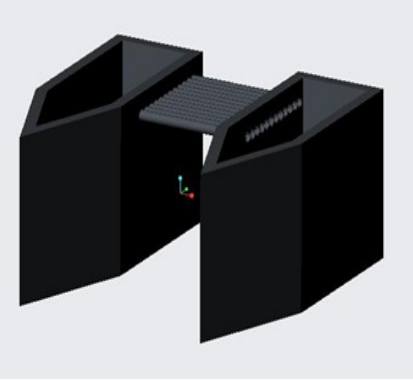


SPECIFICATIONS

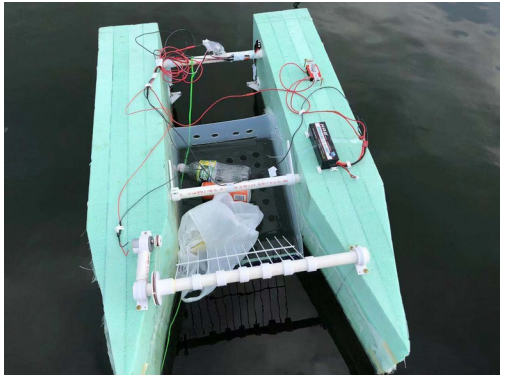
Impacts

- **Environmental**
  - Reduces microplastic levels
- **Social**
  - Decrease in river cleaning workers.
- **Ethical**
  - Raises sense of environmental protection.
- **Technical**
  - New method of river trash collection.

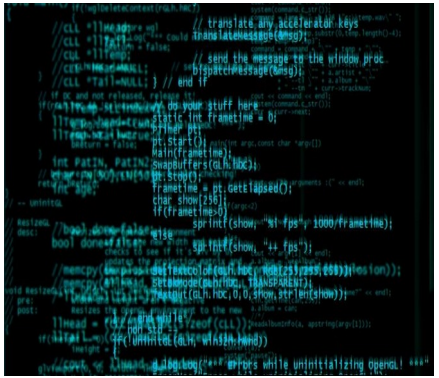
DESIGN PROGRESS



Fall Term 2018 - Platform Design

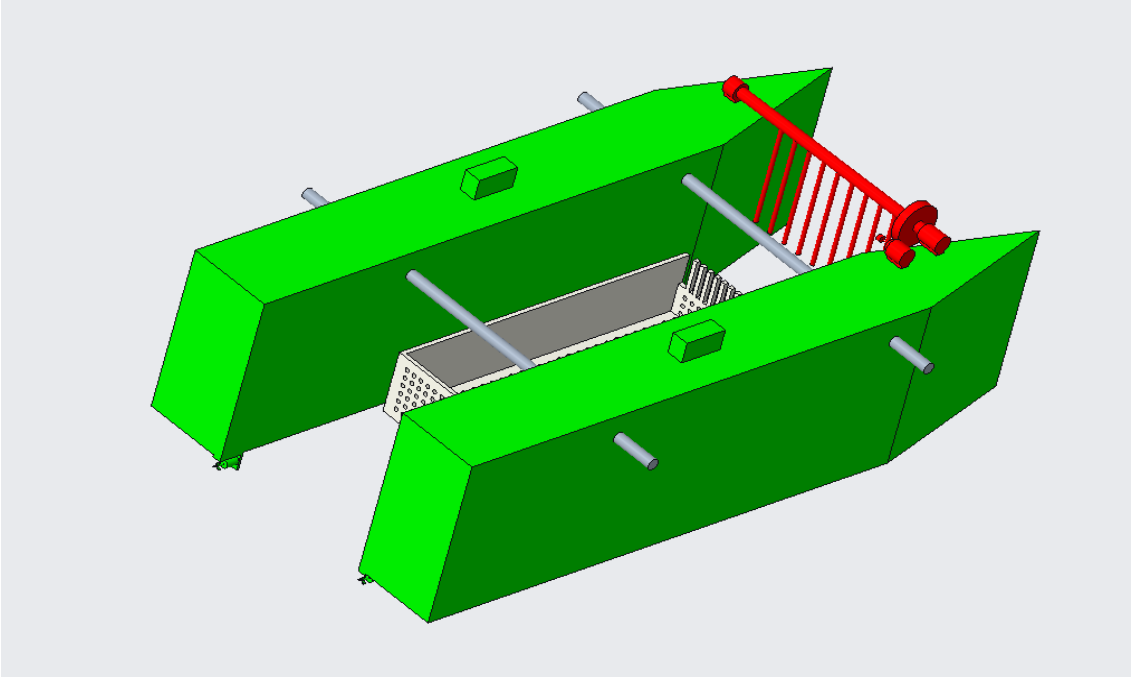


Winter Term 2018 - Device on board and basic functionalities



Spring Term 2019 Automation and other improvements

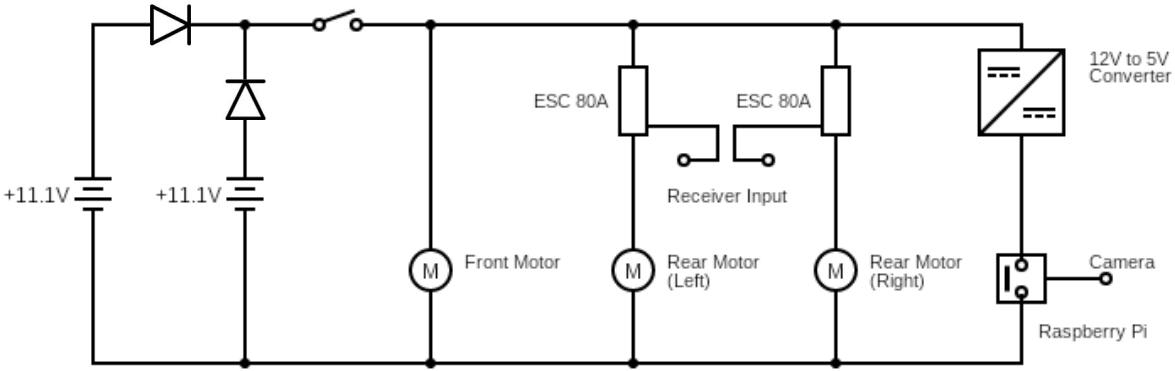
CREO Design For Prototype



Questions to address for “final” product

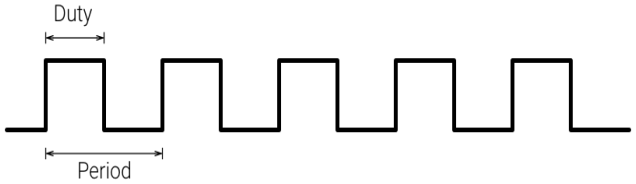
- **Fine Tuning**
  - Fixing problems from previous term.
- **Find out problems before actual product**
  - Find unpractical parts, find conflicts.
- **Create a platform for experiment**
  - Test power system, collection system and other parts.

CIRCUIT DESIGNS



Circuit Schematic

Pulse = 1ms (off) to 2ms (full power)



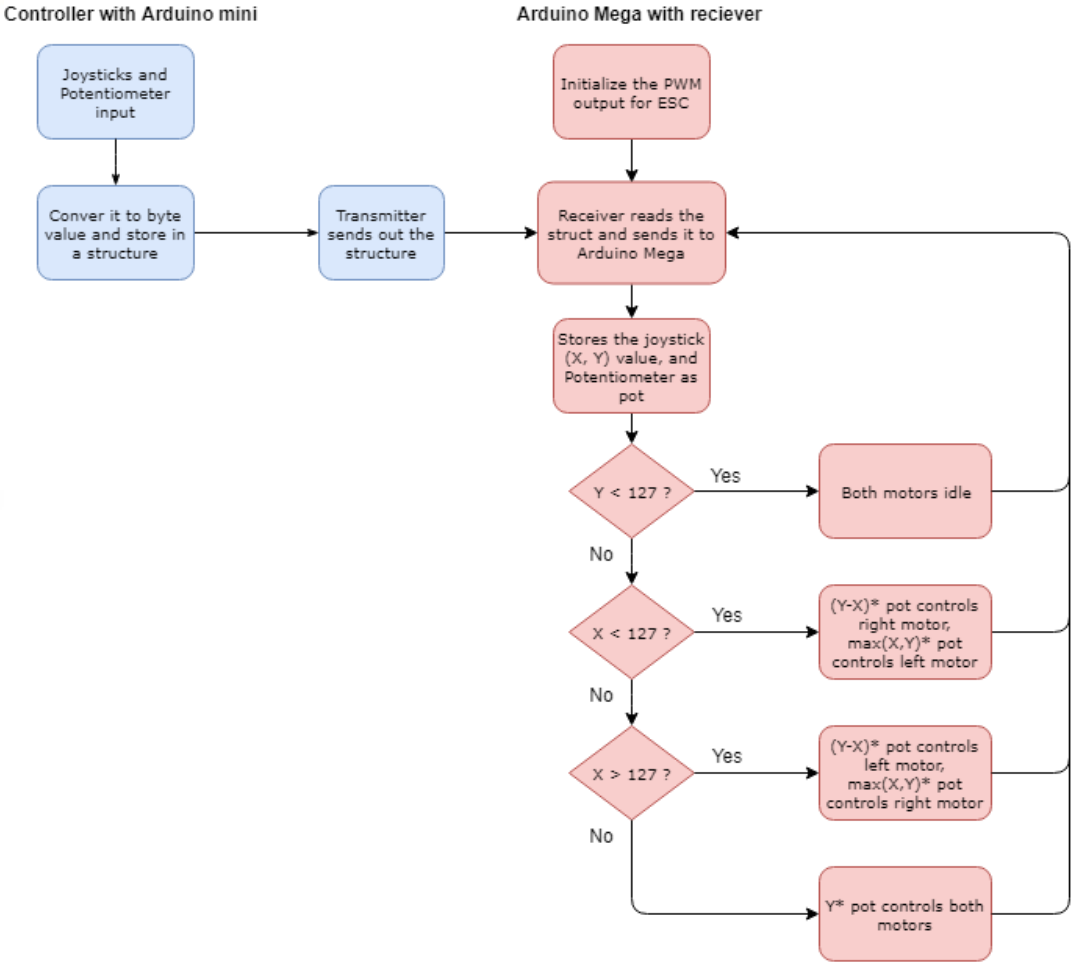
Frequency = 50 Hz

PWM signal
(Receiver Input)

Battery: 6Ah
Rear motor: 24N
Front motor: 30N*cm, 60 RPM

Programming algorithm

Language: Arduino language in Arduino Integrated Development Environment



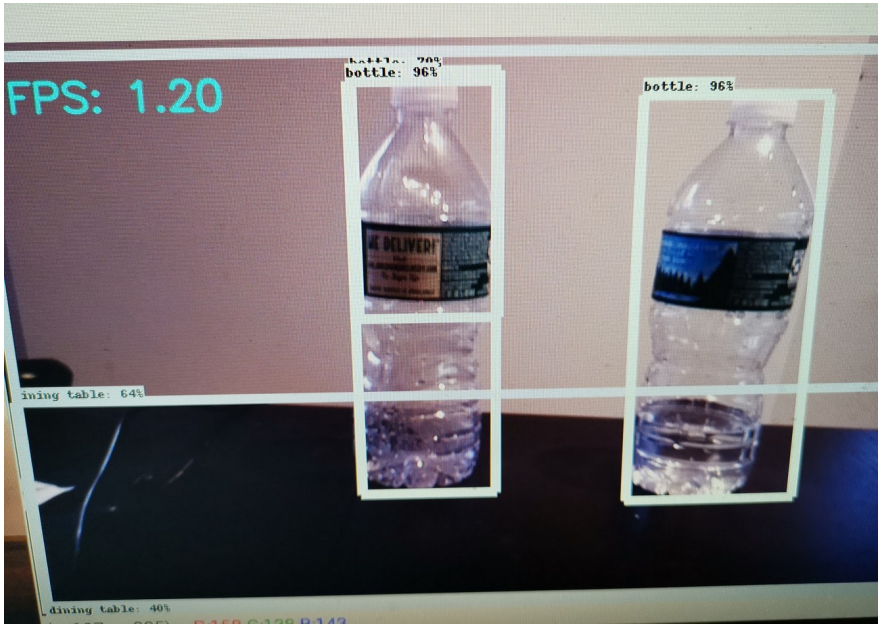
Movement & Trash Collection Experiment



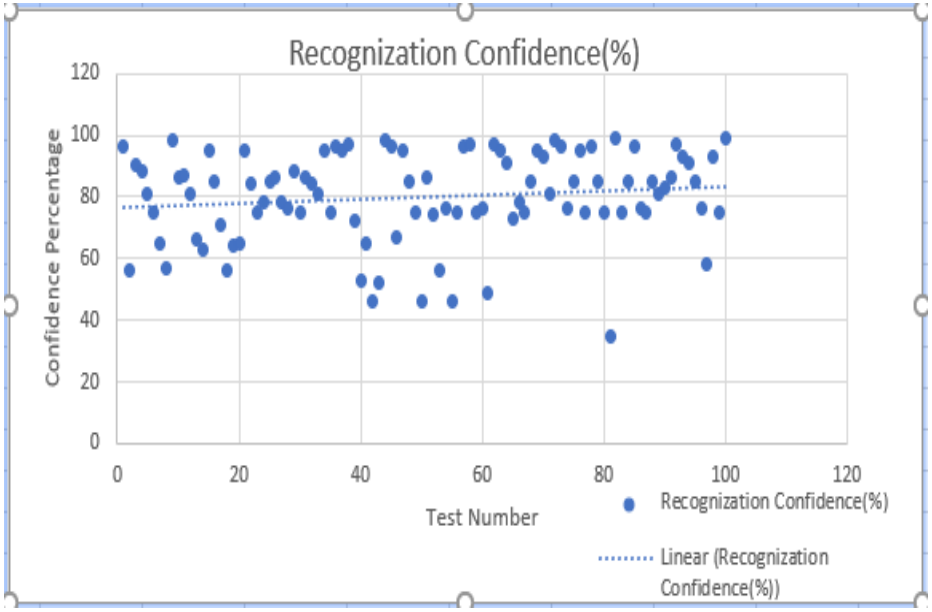


Object Detection

Bottle Detection



Bottle Recognization Experiment



Classes Utilized

- **MEM427- Finite Element Methods**
  - **Finite Element Analysis**
- **MEM201- Fundamentals of Computer Aided Design**
- **Proj101- Project Management For All**
  - **Project Schedule**
  - **Financial Tracking**
  - **Time Management**

Research Sources

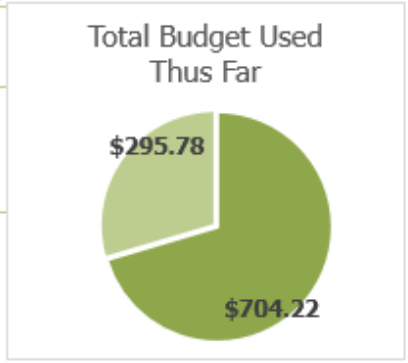
- Search Engine: Google (throughout the project)
- Drexel Library Site (In the beginning)
- Youtube Tutorials (throughout the project)



COST ANALYSIS

RIVER CLEANER BUDGET

BUDGETED AMOUNT	TOTAL COSTS	DIFFERENCE
$1,000.00	$704.22	$295.78





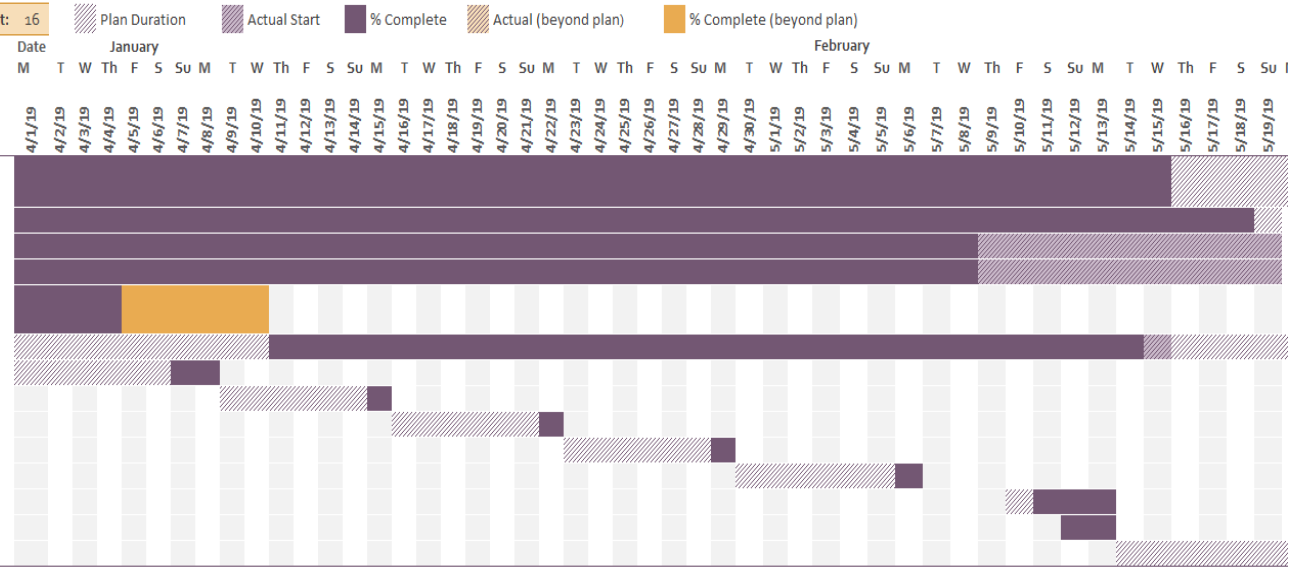
Project Schedule

Automatic River Cleaner

Period Highlight: 16

Plan Duration | Actual Start | % Complete | Actual (beyond plan) | % Complete (beyond plan)

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Material Procurement	11/15	240	10/18/2018	210	100%
Construction	1/12	128	1/14/2019	125	100%
Testing	1/25	115	1/30/2019	110	90%
Troubleshooting	1/25	115	2/6/2019	103	90%
Equation Conceptualization	3/22	14	3/22/2019	20	100%
Programming	3/22	60	4/11/2019	35	99%
Weekly Report 1	4/1	8	4/7/2019	2	100%
Weekly Report 2	4/9	7	4/15/2019	1	100%
Weekly Report 3	4/16	7	4/22/2019	1	100%
Weekly Report 4	4/23	7	4/29/2019	1	100%
Weekly Report 5	4/30	7	5/6/2019	1	100%
Final Report	5/10	4	5/11/2019	3	100%
Standards	5/12	2	5/12/2019	2	100%
Final Presentation	5/14	7	-	-	0%



FUTURE WORK

- Aesthetics
- Control

  Automation
- Energy

  Solar panel integration

What We Learned

- Always have a backup plan
- Explore all possible avenues
- Tolerance is **KEY**





References

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- Thompson, Avery. “You Could Drive This River-Cleaning Trashbot From Anywhere.” Popular Mechanics， Popular Mechanics, 3 Apr. 2018, www.popularmechanics.com/technology/robots/a19676406/chicago-river-trashbot-urban-rivers/
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